

Bioinspired Durable Antibacterial and Antifouling Coatings Based on Borneol Fluorinated Polymers: Demonstrating Direct Evidence of Antiadhesion

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Cite This: *ACS Appl. Mater. Interfaces* 2021, 13, 33417–33426



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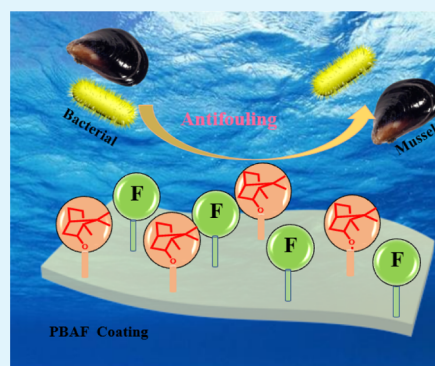
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Supporting Information

ABSTRACT: Substituting natural products for traditional poison-killing antifouling agents is an efficient and promising method to alleviate the increasingly serious ecological crisis and aggravate the loss due to marine biofouling. Herein, the successful synthesis of poly(methyl methacrylate-co-ethyl acrylate-co-hexafluorobutyl methacrylate-co-isobornyl methacrylate) copolymer (PBAF) with borneol monomers and fluorine by a free radical polymerization method is reported. The PBA_{0.09}F coating exhibits outstanding antibacterial and antifouling activity, achieving 98.2% and 92.3% resistance to *Escherichia coli* and *Staphylococcus aureus*, respectively, and the number of *Halophora* sp. adhesion is only 26 (0.1645 mm²) in 24 h. This remarkable antibacterial and antifouling performance is attributed to the incorporation of fluorine components into the copolymer, which induces a low surface energy and hydrophobicity and the complex molecular structure of the natural nontoxic antifouling agent borneol. In addition, the results showed that the contents of the adhesion-related proteins mfp-3, mfp-5, and mfp-6 were significantly reduced, which proved that natural substances affect the secretion of biological proteins. Importantly, the PBAF coating exhibits excellent environmental friendliness and long-term stability. The antifouling mechanism is clarified, and an effective guide for an environmentally friendly antifouling coating design is proposed.

KEYWORDS: antibacterial, antifouling, biologically inspired, natural isobornyl, fluorine



1. INTRODUCTION

Bacteria, algae, barnacles, and mollusks (marine organisms) grow and settle on the surface of ships, pipes, fishing nets, and bridge posts, which are immersed in the sea.^{1–3} The attached biofilm colonies reduce the maneuverability of ships and increase roughness, fuel consumption, economic loss, and biological invasion, thus posing an urgent challenge to marine antifouling.^{4–6} Currently, a variety of biological, physical, and chemical antifouling technologies have been developed to prevent the adhesion of marine biofouling.^{7,8} Among them, antifouling coatings are undoubtedly the easiest to implement and are a cost-effective method.^{9,10} Traditional antifouling coatings, such as tributyltin (TBT) and copper-based coatings, depend on the release of antifouling agents. However, the long-lasting toxicity of antifouling agents seriously jeopardizes marine organisms and the ecological environment; these agents were banned by the International Marine Organization (IMO) in 2008.^{11–13} Therefore, there is an urgent need to design and develop environmentally friendly and nontoxic or low-toxicity antifouling coatings.

Currently, polyethylene glycol (PEG), zwitterions, hydrogels, cationic polymers, and fluoropolymers as nontoxic antifouling coatings have received special attention^{14–16}

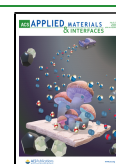
owing to their unique surface properties, for example, surface energy and wettability. As a representative, fluorinated polymers^{17–19} have excellent self-cleaning, corrosion resistance, chemical resistance, and barrier properties, proposing bright prospects in marine antifouling. Undoubtedly, fluorine elements are evenly and orderly dispersed on the polymer surface to achieve excellent antifouling properties because of the low surface energy, and biofouling can be desorbed only by washing or shaking with seawater.^{20,21} In addition, fluorinated polymers also possess excellent mechanical properties,^{22,23} such as outstanding tear strength and superb puncture resistance, making them important coating materials.

In addition to mechanical, chemical stability, and film-forming properties, the selection of antifouling coating materials should consider cost-effectiveness and the low-toxicity control of biofouling. Simultaneously, the first step of

Received: April 1, 2021

Accepted: June 28, 2021

Published: July 12, 2021



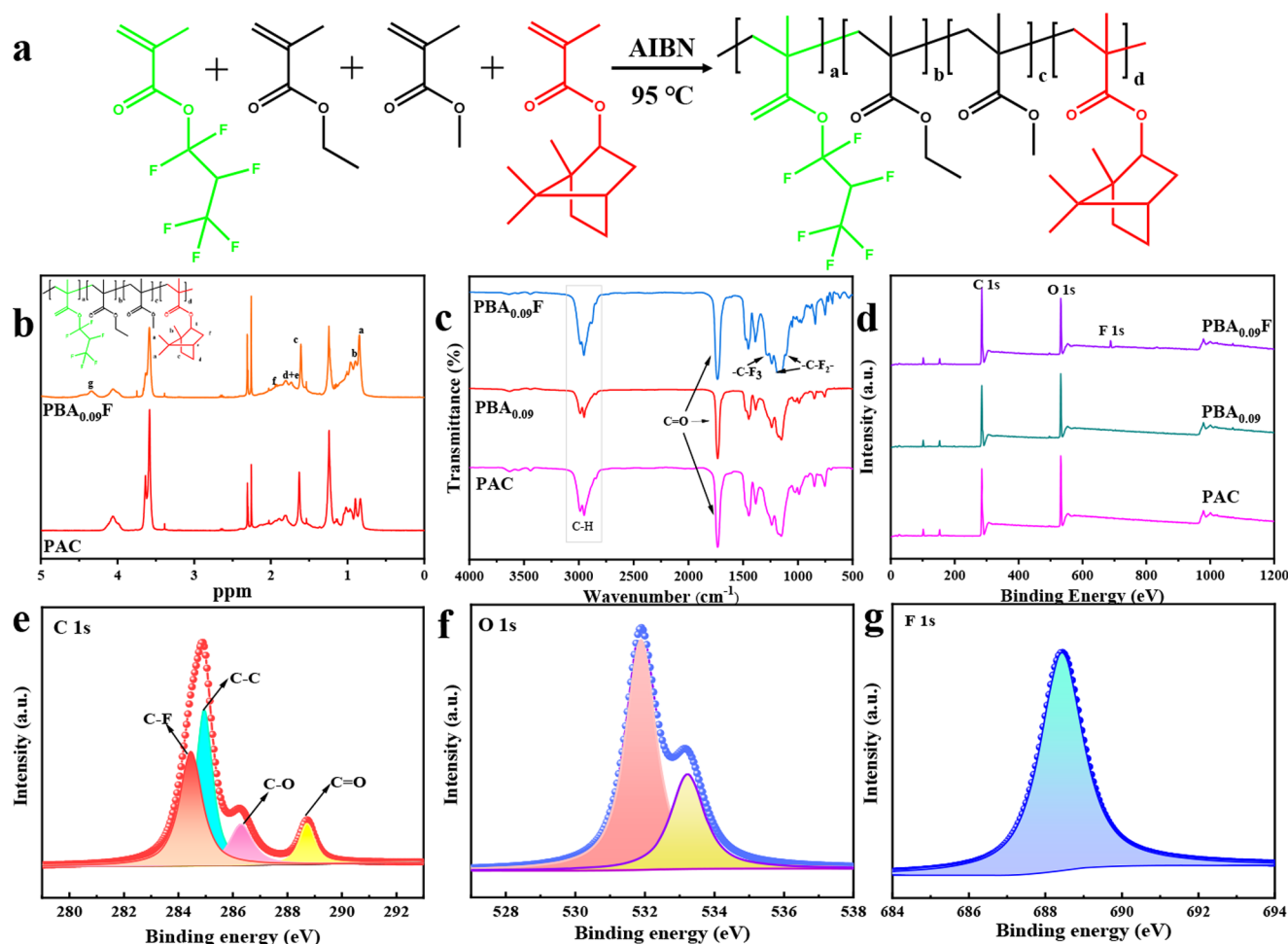


Figure 1. (a) Reaction equation. (b) ¹H NMR spectra of the PAC and PBA_{0.09}F. (c) FTIR spectra of PAC, PBA_{0.09}F, and PBA_{0.09}F. (d) XPS spectra survey of PAC, PBA_{0.09}F, and PBA_{0.09}F. (e) C 1s, (f) O 1s, and (g) F 1s of the PBA_{0.09}F.

biofouling attachment is the rapid deposition of microorganisms (proteins and bacteria) to form biofilms, and then the adhesion of large fouling organisms will occur.^{24–26} Therefore, in the early stage of fouling, effectively inhibiting and improving the process of bacterial reproduction and colonization is extremely important and meaningful for antifouling applications.^{27–29} Generally, antiadhesion of bacteria can be achieved by repelling or killing contact bacteria.³⁰ To uphold the principle of environmental friendliness, natural or bionic strategies should be adopted as much as possible to expel bacterial adhesion.

Borneol,^{31–33} a natural and effective Chinese herbal medicine, can be extracted from many medicinal plants, including lavender, valerian, and chamomile. It has a unique structure of crystalline cyclic alcohol with two enantiomeric forms and possesses a variety of biomedical functions, including anti-injury, analgesic, and anti-inflammatory. More importantly, borneol based on natural herbs has an excellent broad-spectrum antibacterial ability. Borneol-based polymer, isobornyl methacrylate (IBOMA),^{34,35} exhibits resistance to bacterial adhesion, high rigidity, and good thermal stability. According to the above, borneol-based polymers can be used to make biocompatible antibacterial coatings.

Herein, we designed a new type of antibacterial and antifouling coating by isobornyl methacrylate and fluorine-containing polymers. The poly(methyl methacrylate-*co*-ethyl

acrylate-*co*-hexafluorobutyl methacrylate-*co*-isobornyl methacrylate) copolymer (PBAF) coatings were synthesized by radical polymerization that comprised different contents of isobornyl methacrylate and fluorine monomers. The influence of different components on the antibacterial and antifouling performance was investigated, and the antibacterial and antifouling mechanisms were analyzed. Additionally, the environmental influence of coatings was assessed by the growth curve of bacteria and algae. This research provides more possibilities for marine antifouling coatings.

2. EXPERIMENTAL SECTION

2.1. Synthesis of PBA_{0.09}F. First, 60 g of xylene (0.57 mol) and 30 g of propylene glycol methyl ether (PGM) (0.33 mol) were added to a four-necked flask via insertion with a thermometer, condenser, stirrer, and dropping funnel, and then the mixture was heated to 95 °C. Next, 47 g of methyl methacrylate (MMA) (0.47 mol), 23 g of ethyl acrylate (EA) (0.23 mol), 20 g of isobornyl methacrylate (IBOMA) (0.09 mol), 10 g of 2,2,3,4,4,4-hexafluorobutyl methacrylate (HFBMA) (0.04 mol), and 1.2 g of 2,2'-azobis(2-methylpropionitrile) (AIBN) were dissolved in a 30 g mixture of xylene and PGM (the mass ratio was 2:1). The above mixed solution was dropped into a three-necked flask within 5 h at 95 °C under nitrogen atmosphere protection. After the addition was complete, another 0.3 g of benzoyl peroxide (BPO) was dissolved in 5 g of propylene glycol methyl ether solution, and the mixture was added to the four-necked flask. The temperature was maintained for 1.5 h, and

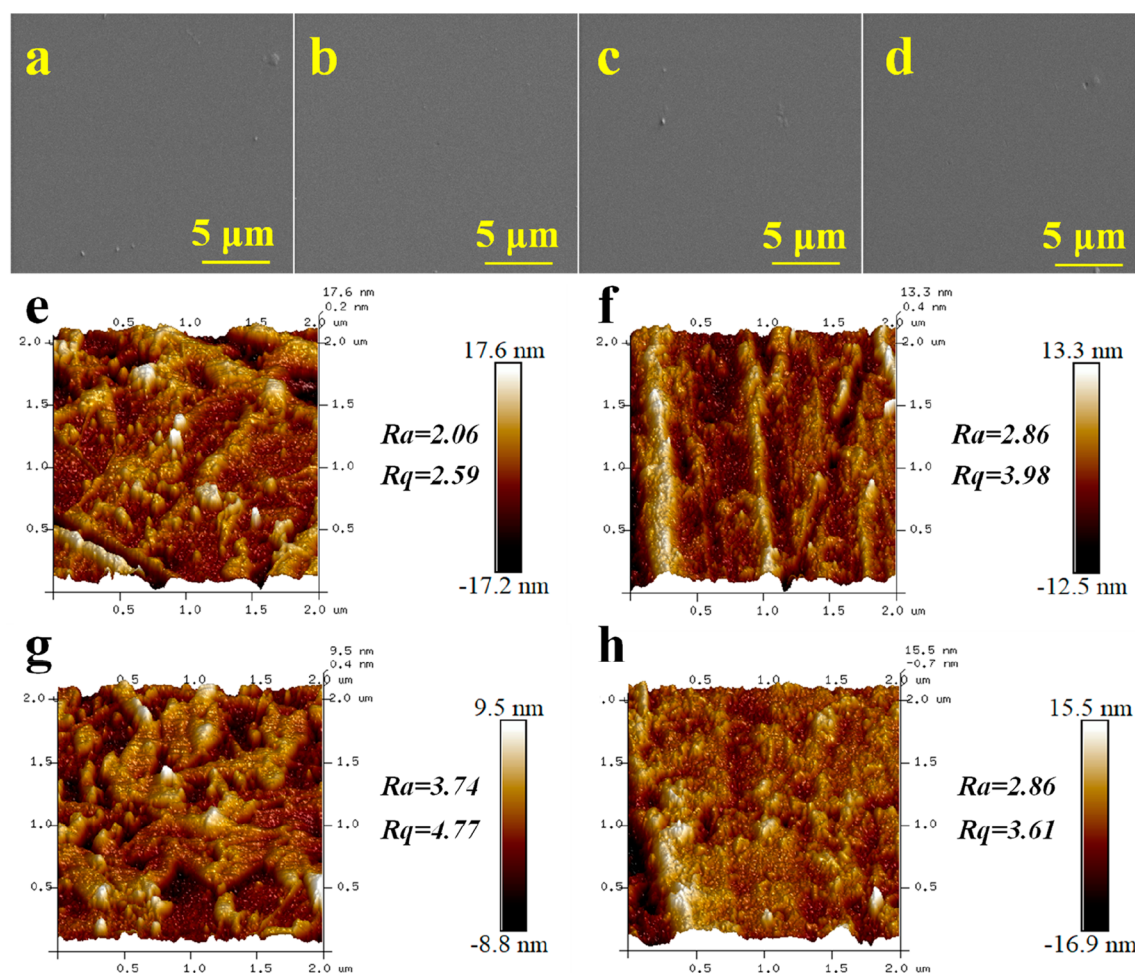


Figure 2. SEM images of the (a) PAC and (c) PBA_{0.09}F coatings before and (b) PAC and (d) PBA_{0.09}F coatings after immersion in artificial seawater for 14 days. AFM images of the (e) PAC and (f) PBA_{0.09}F coatings before and (g) PAC and (h) PBA_{0.09}F coatings after immersion in artificial seawater for 14 days.

then the mixture was cooled to room temperature to obtain the PBA_{0.09}F copolymer. Except for the different monomers, other PBAFs were synthesized in the same way.

2.2. Preparation of the PBAF Film. First, different substrate materials, such as stainless steel sheets (25 × 15 mm) and glass sheets (25.4 × 76.2 mm), were cleaned with ethanol before coating. Second, the copolymer was dropped onto the surface of the base material and coated at a rotation speed of 300 rpm. Finally, the coated samples were dried in an oven at 80 °C for later use.

The polymer was synthesized by free radical polymerization, as shown in Figure 1a, denoted as PBA_xF ($x = 0.045, 0.065, \text{ and } 0.09$) according to the mass percentage of different IBOMA monomers (Table S1). The components without IBOMA monomers were named PAC and PACF. The yields of different copolymers and the solid contents of different copolymers can be found in Table S2. The characterization and detailed antifouling performance test are listed in the Supporting Information.

3. RESULTS AND DISCUSSION

3.1. Structure and Characteristics of the PBAF. The composition and molecular structure of PBAF and PAC copolymers were determined by proton nuclear magnetic resonance (¹H NMR) spectra, as presented in Figure 1b. The peak at $\delta = 3.61$ is assigned to the methyl ester group. For copolymer PBA_{0.09}F, new chemical shifts at 4.42 and 1.94 ppm correspond to the $-\text{CH}$ and $-\text{CH}_2-$ of the isobornyl group,³⁶

respectively, which indicate that the PBA_{0.09}F copolymer was synthesized successfully.

The molecular weights and polydispersity of the PBAF copolymers were measured by gel permeation chromatography (GPC), as shown in Table S2. The molecular weight (Mw) distribution of the copolymers was between 3×10^4 and 4×10^4 , and the polydispersity index (PDI) was concentrated from 1.63 to 1.90. This minute variance is mainly due to the different molecular weights of the reactive monomers, which slightly affect subsequent performance.

To determine the chemical composition and local structure, Fourier transform infrared (FTIR) and X-ray photoelectron spectroscopy (XPS) analyses were performed. The infrared spectra of PAC, PBA_{0.09}, and PBA_{0.09}F are shown in Figure 1c. For all samples, the characteristic peaks in the region between 2800 and 3000 cm^{-1} are classified as the methyl and methylene absorption peaks in MMA and EA. Compared to that of PAC, the absence of any specific peaks and the absorption peak at 1630 cm^{-1} (the $\text{C}=\text{C}$ bond absorption peak) in the spectra of PBA_{0.09} indicated that no residual monomer was present and that the reaction proceeded completely. The absorption peak at approximately 1286 cm^{-1} is attributed to the $\text{C}-\text{F}_3$ bond stretching vibration, whereas the peaks at 1096 and 1135 cm^{-1} are assigned to the $\text{C}-\text{F}_2$ bond stretching vibration in copolymer PBA_{0.09}F. The results are consistent with the ¹H

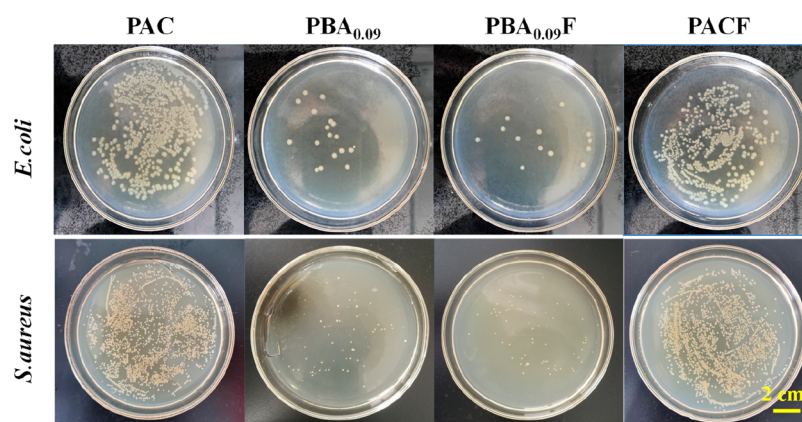


Figure 3. Antibacterial image of different PBAF coatings for *E. coli* and *S. aureus*.

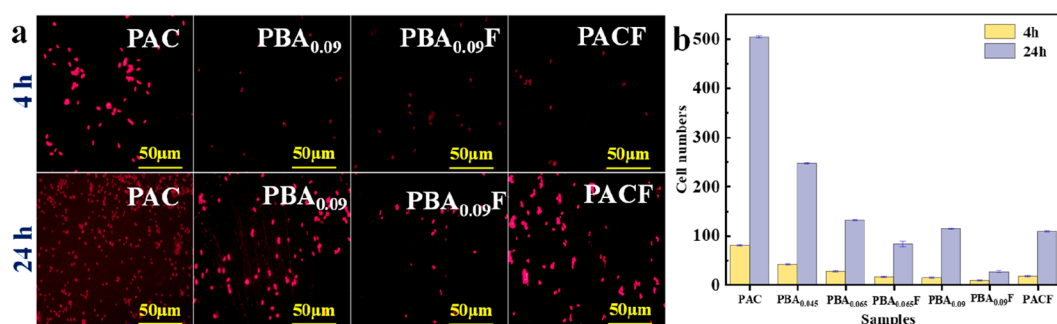


Figure 4. Fluorescence microscopy images of the different PBAF coatings immersed in *Halamphora sp.* algae solution after 4 and 24 h. (b) The number of *Halamphora sp.* attached to the surface of different coatings.

NMR spectrum, demonstrating that copolymer PBAF was successfully synthesized. The XPS of PAC, PBA_{0.09}, and PBA_{0.09}F was also carried out, and the survey spectra are shown in Figure 1d. All samples contained C and O elements, whereas F appeared only in PBA_{0.09}F. In the C 1s high-resolution spectrum of PBA_{0.09}F (Figure 1e), four characteristic peaks located at 284.5, 285, 286.3, and 288.7 eV can be observed, which correspond to C–F, C–C, C–O, and C=O bonds, respectively.³⁷ Two peaks at 531.8 and 533.2 eV in the O 1s spectrum (Figure 1f) are identified as C–O and C=O bonds, respectively. In addition, the peak at a 688.4 eV binding energy can be ascribed to the C–F bond (Figure 1g), demonstrating the existence of F in PBA_{0.09}F and the successful aggregation.

3.2. Morphology of the PBAF Coatings. The morphology and roughness of the coating surface have a critical influence on the adhesion of biofouling (especially diatoms).³⁸ A rough or uneven surface can provide more attachment points for biofouling. In addition, the higher roughness will promote the “rooting” of the adhesive secreted by biofouling on the surface of the material and increase the bonding strength. To investigate the surface morphologies and roughness of the PAC and PBA_{0.09}F coatings, scanning electron microscopy (SEM) and atomic force microscopy (AFM) images of the coatings on the substrates were recorded.

The SEM images of the PAC and PBA_{0.09}F coatings before and after immersion in artificial seawater for 14 days are presented in Figure 2a–d. The surfaces of unsoaked PAC and PBA_{0.09}F are relatively smooth and flat without wrinkles, indicating that both coatings have good film-forming properties. Surprisingly, compared to that of the unsoaked coatings, the morphology of the PAC and PBA_{0.09}F coating surfaces did

not change, maintaining good smoothness after being immersed in water for 14 days. Furthermore, AFM was implemented to digitally evaluate the surface roughness of the coating layers before and after immersion in artificial seawater, and the results are shown in Figure 2e–h. R_a and R_q were used to measure roughness, and their values did not change much before and after immersion of the PAC coating and PBA_{0.09}F coating in artificial seawater. The R_a and R_q of the digitized PBA_{0.09}F coating before immersion are 2.86 and 3.90 nm, respectively, whereas after immersion in water for 14 days, the values decreased slightly to 2.81 and 3.61 nm, respectively. This small decrease in the R_a and R_q values before and after immersion indicates that the surface roughness of the coating is hard enough to guarantee a long service life without significant changes. The SEM and AFM characterization results show that the PBA_{0.09}F coating can maintain good stability in artificial seawater, which promises its long-term antifouling application under complex marine conditions.

3.3. Biofouling Tests of the PBAF Coatings. To evaluate the marine antifouling performance of the coatings, the adhesion behavior of bacteria, diatoms, and mussels was studied. As coatings will be used in the marine environment, marine bacteria were first studied as an antibacterial adhesion model. As displayed in Figure S1, the largest number of colonies was observed for the PAC coating in comparison to those of other coatings which had fewer colonies, indicating that the PAC coating has poor antibacterial properties and that the PBAF coating has certain antibacterial adhesion properties. The antibacterial pictures of marine bacteria cannot quantify the specific properties of other coatings with different components, so Galanz-positive bacteria (*Staphylococcus*

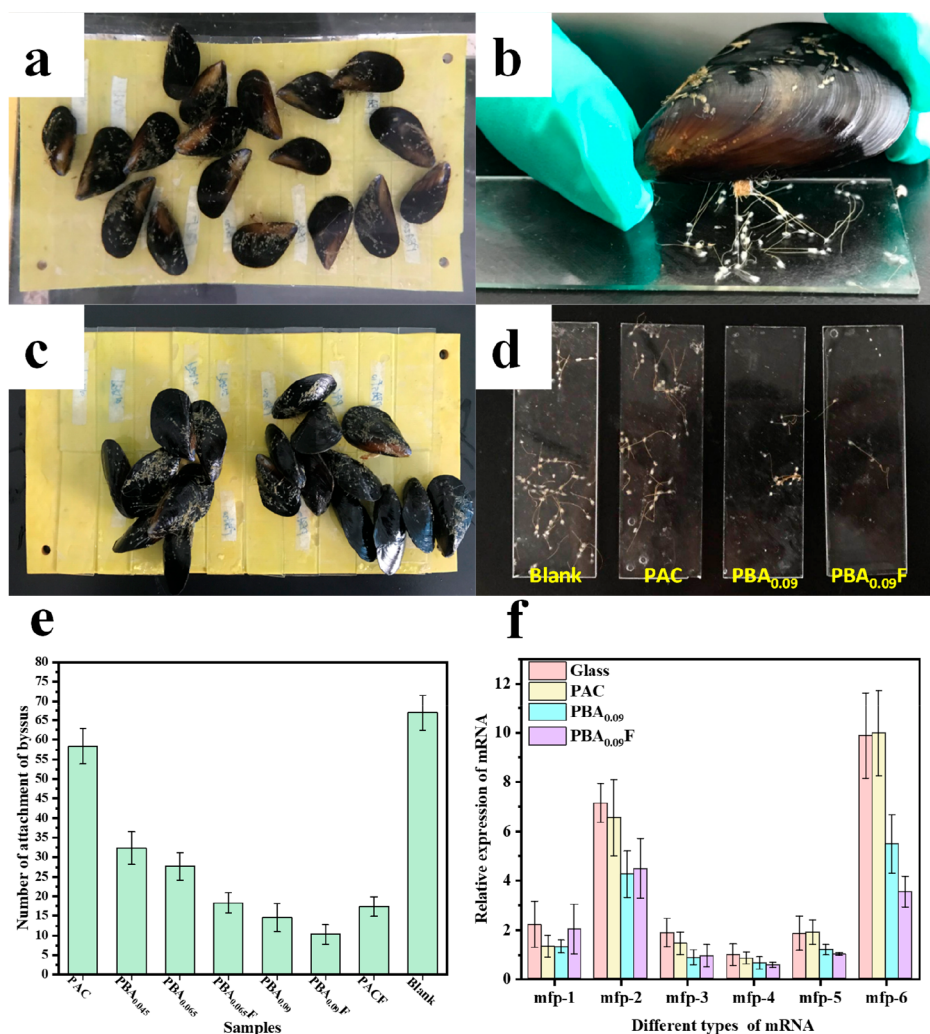


Figure 5. (a) Image of mussels randomly scattered on the surface of the samples. (b) Image of mussels attached to the surface of the sample after 24 h. (c) Mussels attached to glass sheets and (d) byssus plates on the blank, PAC, PBA_{0.09}, and PBA_{0.09}F samples. (e) Average number of byssus plates on different samples in three experiments. (f) Relative expression level of adhesion protein-related RNA in glass, PAC, PBA_{0.09}, and PBA_{0.09}F samples.

aureus) and Galanz-negative bacteria (*Escherichia coli*) were used as typical bacteria to evaluate antibacterial performance. As shown in Figure 3, consistent with the results of marine bacteria, the PAC coating has a low antibacterial rate, whereas PBAF coatings with different contents show high antibacterial efficiency against *E. coli* and *S. aureus* within 24 h owing to the excellent antibacterial activity of isoborneol. Additionally, the weak antibacterial performance of the PACF coating indicates the weak antibacterial properties of fluorine. The antibacterial pictures of other components are shown in Figure S2. Specifically, the antibacterial rate of *E. coli* was recorded to be 80.4%, 97.1%, 95.9%, 97.6%, and 18.1%, and the antibacterial rates of *S. aureus* were 39.2%, 77.4%, 77.2%, 91.8%, and 3.2% for PBA_{0.045}, PBA_{0.065}, PBA_{0.065}F, PBA_{0.09}, and PACF, respectively. The highest antibacterial rates of the PBA_{0.09}F coating against *E. coli* and *S. aureus* were 98.2% and 92.3%, respectively, indicating that the antibacterial activities against Gram-negative bacteria were slightly higher than those against Gram-positive bacteria (Figure S3).

To simulate actual marine applications, stainless steel plates as the base of *Halophora sp.* and *Nitzschia closterium* with a strong adhesion as typical diatoms were applied to evaluate

algae adhesion ability. The coatings were immersed in a certain concentration of *Halophora sp.* liquid and then removed at the 4th and 24th hours to observe the amount of attached seaweed. As shown in Figure 4 and Figure S4, the coatings with the IBOMA monomer and HFBMA monomer display better resistance to diatom adhesion than the PAC coating in the first 4 h of attachment. The data show that the number of attachments of the PAC, PBA_{0.045}, PBA_{0.065}, PBA_{0.065}F, PBA_{0.09}, PBA_{0.09}F, and PACF are 79, 40, 27, 15, 13, 8, and 16 (0.1645 mm²), respectively. It is undeniable that the number of attachments gradually increases over time. Compared to the first 4 h, the number of attachments of PAC coating increased by 6.38 times at 24 h. Delightfully, the PBA_{0.09}F coating exhibited excellent antidiatom adhesion performance after 24 h with only 26 diatoms adhesion, which is far less than the number of algae attached to the PAC coating after 4 h. The coatings containing the IBOMA monomer all exhibited good antialgae adhesion performance; the antidiatom adhesion rate is positively correlated with an increase in the IBOMA monomer content. Simultaneously, the amount of antidiatom adhesion of the PACF coating is much less than that of the PAC coating, which proves that the HFBMA monomer

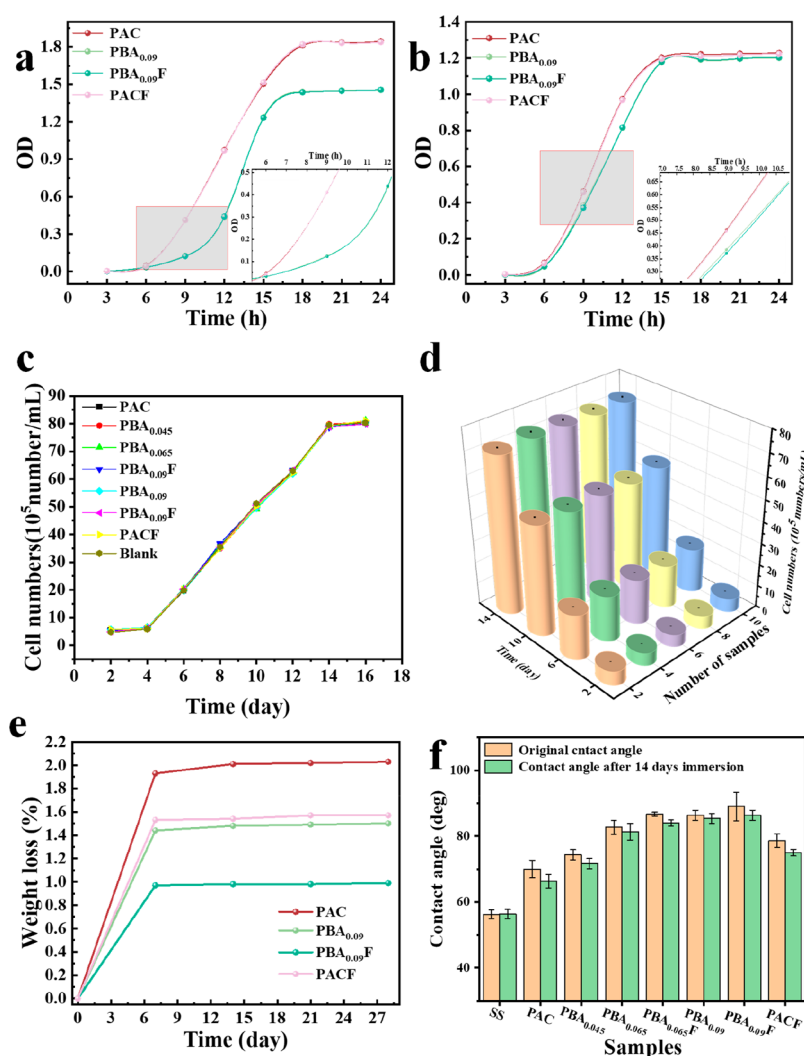


Figure 6. OD₆₀₀ test of (a) *E. coli* and (b) *S. aureus* versus different samples. (c) Growth curve of *Nitzschia closterium* versus different samples. (d) Growth curve of *Nitzschia closterium* after placing different numbers of PBA_{0.09}F samples. (e) Immersion weight loss test of different samples. (f) Contact angle of coated surfaces before and after immersion in artificial seawater for 14 days.

certainly hinders diatom adhesion. In addition, the attachment of *Halamphora* sp. and *Nitzschia closterium* crescents after 14 days were also investigated (Figure S5). A large number of agglomerated diatoms appeared on the surface of the coatings. In agreement with the previous results, the number of seaweeds attached to the surface of the PBA_{0.09}F coating was the lowest.

Large biofouling, especially hard shells, is the most harmful in practical applications. To systematically investigate the antifouling performance of the prepared coatings, a mussel settlement assay was carried out. As seen in Figure 5a, the mussels were randomly placed on the surfaces before the test. However, most of the mussels moved and reunited after 24 h (Figure 5b). The byssuses of the mussels attached to each sample were carefully cut off, and then the byssus plates on the coated surfaces were counted. Figure 5c,d shows the byssus plate attached to the blank and the coated glass sheets. As shown in Figure 5e, compared to the blank glass sheet, the glass sheet with PAC coating has almost no ability to resist mussel adhesion. As the IBOMA monomer content was increased, the byssus plates attached to the coated surfaces began to show a downward trend. Significantly, the coating's ability to resist mussel adhesion is further enhanced by adding

HFBMA monomer. It can be observed from the experimental results that the PBA_{0.09}F coating has the best resistance to mussel adhesion.

To further seek the reasons for the antiadhesion of PBA_{0.09}F coating, the quantitative real-time PCR (RT-qPCR, ABI Prism 7500) analysis was carried out. It is well-known that the adhesion plaque of mussel is a part of the mussel foot, which is the "sucker" connecting the substrate material of mussels. There are six proteins in adhesive plaque (mussel foot proteins-1, -2, -3, -4, -5, and -6). Among them, mussel foot proteins (mfp)-2, -3, -4, -5, and -6 are secreted by phenolic glands. mfp-3, -5, and -6 at the adhesion interface play a major role in adhesion.^{41,42} As shown in Figure 5f, there was some significant difference in the expression of the related regulatory genes of all mfp's on the glass and PBA_{0.09}F materials, which means the expression level on the PBAF surface is lower than that of the glass. The related regulatory genes of mfp-3, -5, and -6 on the glass surface are 1.99, 1.74, and 2.8 times that of PBA_{0.09}F, respectively. These results suggest that some components in PBA_{0.09}F affect protein secretion and gene expression. To further explain which component is the dominant driving force, the protein expression on PAC and PBA_{0.09} coatings was evaluated. There were significant

differences in the expression of key adhesion proteins (mfp-3, -5, and -6) between PAC and PBA_{0.09}, indicating that the presence of the IBOMA monomer could effectively reduce the expression of the adhesion protein in mussel. Combined with previous reports, mussels can identify the external bacterial membrane to determine whether it is deposited on the surface.^{43–45} We speculate that the synergistic effect of IBOMA and HFBMA in PBAF coating makes it difficult for bacteria to form bacterial films on the surface of the coating. Mussels found that there was no bacterial membrane on the surface of the PBAF coating, which was not conducive to adhesion. Therefore, the expression of mfp-3, mfp-5, and mfp-6 decreased in vivo.

3.4. Antifouling Activity of the PBAF Coatings. The overall antifouling test results infer that the PBA_{0.09}F coating has the best antifouling performance, which is attributed to the bactericidal performance of the IBOMA monomer and the low surface energy of the HFBMA monomer. To analyze the influence of each monomer on the antifouling performance, especially the IBOMA monomer, the growth characteristics of bacteria and diatoms were investigated. As shown in Figure 6a,b, compared to the blank and PAC coating, PACF does not affect the growth and reproduction of *E. coli* and *S. aureus*, confirming that the HFBMA monomer is not toxic to bacteria. The weak antibacterial efficiency of PACF is mainly derived from the low surface energy provided by the HFBMA monomer, which is not conducive to bacterial adhesion. However, the *E. coli* OD values of the coating containing the IBOMA monomer show a significant decrease after 7 h of incubation, indicating that the IBOMA monomer has the main effect on bacteria. Unexpectedly, the effect of the IBOMA monomer coating on *S. aureus* is not more obvious than that of *E. coli*. The OD value is less than that of the PAC coating in the range of 6–15 h; however, the OD value of all coatings is significantly reduced after 15 h. This phenomenon also explains why the PBAF coating has better resistance to Galanz-negative bacteria than Galanz-positive bacteria. After 6 h of incubation, the OD values of coatings with different IBOMA monomer contents were obviously different, representing different *E. coli* densities from high to low in the order of blank, PAC, PACF, PBA_{0.045}, PBA_{0.065}, PBA_{0.065}F, PBA_{0.09}F, and PBA_{0.09} (Figure S6). The growth of diatoms was also investigated. Figure 6c shows the diatom growth results for all coatings, which were soaked in *Nitzschia closterium* for 16 days. Compared to the blank (glass), the diatom growth rate in solutions containing IBOMA or HFBMA monomer displays no significant difference in daily incubation. This observation verified that the coatings are nontoxic to algae and are environmentally friendly coatings. In addition, a biological toxicity experiment was performed, and the results are shown in Figure 6d and Figure S7. Two, four, six, eight, and ten pieces of PBA_{0.09}F coatings were added to each culture medium. The number of diatoms in all samples increased with the extension of culture time, and there was no difference at the same culture days. The PBAF coating had no toxicity to the diatom cells, and the results were consistent with previous results.

To explore the surface wettability of PBAF coatings, a static water contact angle (CA) was conducted under similar conditions. The surface structure of IBOMA molecules is more complex than that of MMA and EA and has a larger hydrophobic cage structure. As shown in Figure 6f, the CA of the original steel sheet was 56.2°, which is a hydrophilic surface, and there was no change after 14 days of immersion.

The hydrophobic properties of the coating gradually improved with a gradual increase in the IBOMA content.³⁹ Quantitatively, the CA reached approximately 86° when the IBOMA content was 0.09 mol; simultaneously, increases in the HFBMA composition led to increases in the CA to approximately 78°. The highest CA value of 89° is recorded for the PBA_{0.09}F coating containing the IBOMA monomer and HFBMA monomer. Studies have shown that better hydrophobic properties have a certain inhibitory effect on diatom adhesion. To survey the stability of the coatings, a static CA test was performed on the coatings immersed in artificial seawater for 14 days. Although the CA values were slightly reduced compared to the original CA values before immersion, the overall change was not large, indicating good stability of the PBAF coating in artificial seawater. At the same time, we measured the surface energy of different coatings (Figure S8). With the addition of IBOMA and HFBMA, the surface energy of the coatings decreased, which had a positive effect on the resistance to fouling organisms. In addition, the coating inevitably undergoes slight hydrolysis under artificial seawater environment conditions; therefore, the weight loss of all coatings was also investigated. As shown in Figure 6e, all coatings undergo relatively rapid hydrolysis within 7 days, but the hydrolysis rate is less than 2%, which is almost negligible compared to that of the currently used coatings. The hydrolysis rate of all coatings tends to be flat and insignificant after 7 days. The weight loss ratio of the coatings containing IBOMA and fluorine and coatings containing only IBOMA decreased slower than that of the PAC coating during the immersion period. The reason may be because the IBOMA monomer and HFBMA monomer in the polymer matrix make the resin more hydrophobic. The weight loss of the other coatings is shown in Figure S9. Similarly, we conducted the immersion thinning experiment, and the results are shown in Figure S10. After 14 days of immersion, we found that the thickness of all coatings decreased slightly, but the thickness reduction of all coatings was less than 2%, which was consistent with the weight loss results. This directly proved that the coating remained stable in the artificial seawater environment and had the conditions for application in marine conditions.

On the basis of the above results and the literature,^{36,40} a possible reaction mechanism for synergistic antifouling is described below. For the PAC surface, water is easily absorbed on the surface, and there is no antifouling functional group, which forms a suitable condition for biofouling to attach. The PBAF coating surface became smoother and more hydrophobic because of the fluorine property, forming a barrier between the biofouling and the substrate, which weakened the interaction between the biofouling and the surface, thereby reducing adhesion. In addition, according to the results of this article and literature reports, the complex molecular structure of borneol on the surface of the PBAF coatings can significantly affect cell adhesion and protein adsorption at the interface, causing effective antifouling of the stereochemistry of the polymer surface. Moreover, borneol, as a natural drug, is a nontoxic antifouling agent, and the microrelease in the coating effectively affects the bacterial sensory system, thereby avoiding bacterial adhesion and further fouling adhesion.

4. CONCLUSIONS

In conclusion, a series of antibacterial and antifouling poly(methyl methacrylate-co-ethyl acrylate-co-hexafluorobutyl

methacrylate-co-isobornyl methacrylate) copolymer (PBAF) coatings with different IBOMA monomers and HFBMA monomers were successfully synthesized by a simple free radical polymerization method. The SEM and AFM results show that the surfaces of the PBAF coatings are relatively smooth and flat without wrinkles before and after immersion in water for 14 days. The PBAF coating exhibited outstanding antibacterial and antifouling activity. Specifically, the PBAF disinfects *E. coli* and *S. aureus* at 98.2% and 92.3%, respectively. The mechanisms driving antifouling were systematically investigated. The fluorine property of the PBAF coating makes the surfaces smoother and more hydrophobic. The complex molecular structure and microrelease of borneol monomers repel biofouling. Furthermore, the PBAF coating displayed excellent environmental friendliness and long-term stability, making it suitable for marine environments. Ultimately, this work will promote the research and application of antifouling coatings in various fields.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.1c06030>.

Additional experimental details, experimental formulation of different copolymers, details of different copolymers, antibacterial image of different coatings for marine bacteria, antibacterial image of different coatings for *E. coli* and *S. aureus*, antibacterial rates of different PBAF coatings for *E. coli* and *S. aureus*, fluorescence microscope images of the different PBAF coatings immersed in *Halophora sp* algae solution, fluorescence microscope images of the different PBAF coatings immersed in *Halophora sp* and *Nitzschia closterium* algae solution, OD₆₀₀ test of *E. coli* and *S. aureus*, digital photographs of *Nitzschia closterium* growth at different times, surface energy of different coatings, weightlessness rate of the different coatings in artificial seawater within 28 days, and thickness change of the coating as a function of immersion time in artificial seawater (PDF)

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<https://pubs.acs.org/doi/10.1021/acsami.1c06030>

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the National Key Research and Development Project (2019YFC0312102), Defense Industrial Technology Development Program (JCKY2019604C001), NSFC-Nuclear Technology Innovation Joint Fund (U1967214), Postdoctoral Science Foundation of China (2019M663416), the Fundamental Research Funds of the Central University and Heilongjiang Touyan Innovation Team Program.

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